

This information is controlled by the U.S. Department of Commerce
Export Administration Regulations 15 CFR Parts 730-774 as EAR99 Technology

REVISION HISTORY					
REV	DESCRIPTION	DATE	DESIGNER	CHECKED BY	ENGINEER
1	INITIAL RELEASE		XX	XX	XX

FABRICATION NOTES

DRC'S MUST BE RUN ON THE GERBERS PRIOR TO BUILDING BOARDS

1.) BOARD MATERIAL: (SEE STACKUP BELOW)

- RF35 □ R04003
- FR-4 □ R04350
- 4000-13EP

2.) BOARD STACKUP

- COMPONENTS ON TOP SIDE ONLY □ COMPONENTS ON BOTH SIDES

LAYERS

FINISHED OVERALL THICKNESS:

- .031 +/- .005
- .062 +/- .005
- .093 +/- .005
- .125 +/- .005

3.) COPPER WEIGHT:

EXTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED

INTERNAL LAYERS: (SEE STACKUP BELOW) OUNCE FINISHED

4.) PLATING:

- HOT AIR SOLDER LEVEL (HASL) EXPOSED AREAS WITH LEADED SOLDER FINISH 50uIN MIN - 200uIN MAX.
- FINISH: AS PER IPC ENIG. SPECIFICATION 4522, 2-4 MICRO-INCHES IMMERSION GOLD OVER 120-240 MICRO-INCHES NICKEL
- OMIKRON WHITE TIN DEPOSITION PROCESS OR ANDREW APPROVED EQUIVALENT TIN PLATING SHALL BE 200 TO 250 MICRO INCHES IN THICKNESS.
- STANDARD SPEC.

5.) DRILLING:

UNLESS OTHERWISE SPECIFIED, PLATE ALL HOLES THRU.
SEE HOLE CHART FOR FINISHED HOLE SIZES AND QUANTITY.

6.) HOLE SIZE TOLERANCES SHALL BE:
(SEE HOLE CHART)

7.) SOLDERMASK:

- TOP SIDE
- BOTTOM SIDE
- LIQUID PHOTO-IMAGEABLE SOLDERMASK SHALL BE APPLIED OVER BARE COPPER UNLESS OTHERWISE SPECIFIED
- GREEN
- BLUE
- RED
- BLACK
- MATTE BLACK

8.) SILKSCREEN:

- TOP SIDE
- BOTTOM SIDE
- CHARACTER LEGEND TO BE SCREENED USING NON-CONDUCTIVE EPOXY BASED INK.
- WHITE
- _____

9.) BOW AND TWIST: MAXIMUM ACCEPTABLE IS 0.75% (IPC-TM-650)

10.) FABRICATE IN ACCORDANCE WITH IPC-6011/6012, CLASS 2 (LATEST REVISIONS)

11.) ACCEPTABILITY TO IPC-A-600 (LATEST REVISIONS)

12.) CONDUCTOR INFORMATION:

DRC MUST BE RUN ON GERBERS BEFORE BUILDING BOARDS UNLESS

APPROVAL IS GIVEN IN WRITING.

SMALLEST TRACE: 6 mil

SMALLEST SPACING: 5.91 mil

CONDUCTOR WIDTHS AND SPACING SHALL BE WITHIN +/-20% OF ARTWORK ORIGINALS.

CONTROLLED IMPEDANCE REQUIREMENTS AS CALLED OUT IN CONTROLLED IMPEDANCE NOTES.

13.) REMOVE ALL BURRS AND BREAK SHARP EDGES 0.015 INCHES MAX

14.) VENDOR INFORMATION:

VENDOR WILL PUT THEIR UL LOGO AND DATE CODE IN SILKSCREEN
ON THE COMPONENT SIDE IN THE UNOBSTRUCTED AREA

BOARD CHARACTERISTICS

Copper Layer Count: 6 Board Thickness: 60.87 mils

Board overall dimensions: 9500.00 mils x 3000.00 mils

Min track/spacing: 6.00 mils / 5.91 mils Min hole diameter: 11.81 mils

Copper Finish: ENIG Impedance Control: Yes

Castellated pads: No Plated Board Edge: No

Edge card connectors: No

Layer Name	Type	Material	Thickness (mils)	Color	Epsilon R	Loss Tangent
SST	Top Silk Screen	Not specified	0 mils	Not specified	1	0
PMT	Top Solder Paste		0 mils		1	0
SMT	Top Solder Mask	Not specified	0.3937 mils	Green	3.3	0
LY01	copper		1.37795 mils		1	0
Dielectric	prepreg	2116	5.11811 mils	FR4 natural	4.45	0
LY02	copper		1.37795 mils		1	0
Dielectric	core	DK	16.92913 mils	FR4 natural	4.6	0
LY03	copper		1.37795 mils		1	0
Dielectric	prepreg	7628	7.71654 mils	FR4 natural	4.74	0
LY04	copper		1.37795 mils		1	0
Dielectric	core	DK	16.92913 mils	FR4 natural	4.6	0
LY05	copper		1.37795 mils		1	0
Dielectric	prepreg	2116	5.11811 mils	FR4 natural	4.5	0.02
LY06	copper		1.37795 mils		1	0
SMB	Bottom Solder Mask	Not specified	0.3937 mils	Green	3.3	0
PMB	Bottom Solder Paste		0 mils		1	0
SSB	Bottom Silk Screen	Not specified	0 mils	Not specified	1	0

Controlled Impedance Top and Bottom Layers

1 Top Sig Edge coupled microstrip .17/.15 mm 100 Ohm differential 10%

2 In1 GND

3 In2 PWR

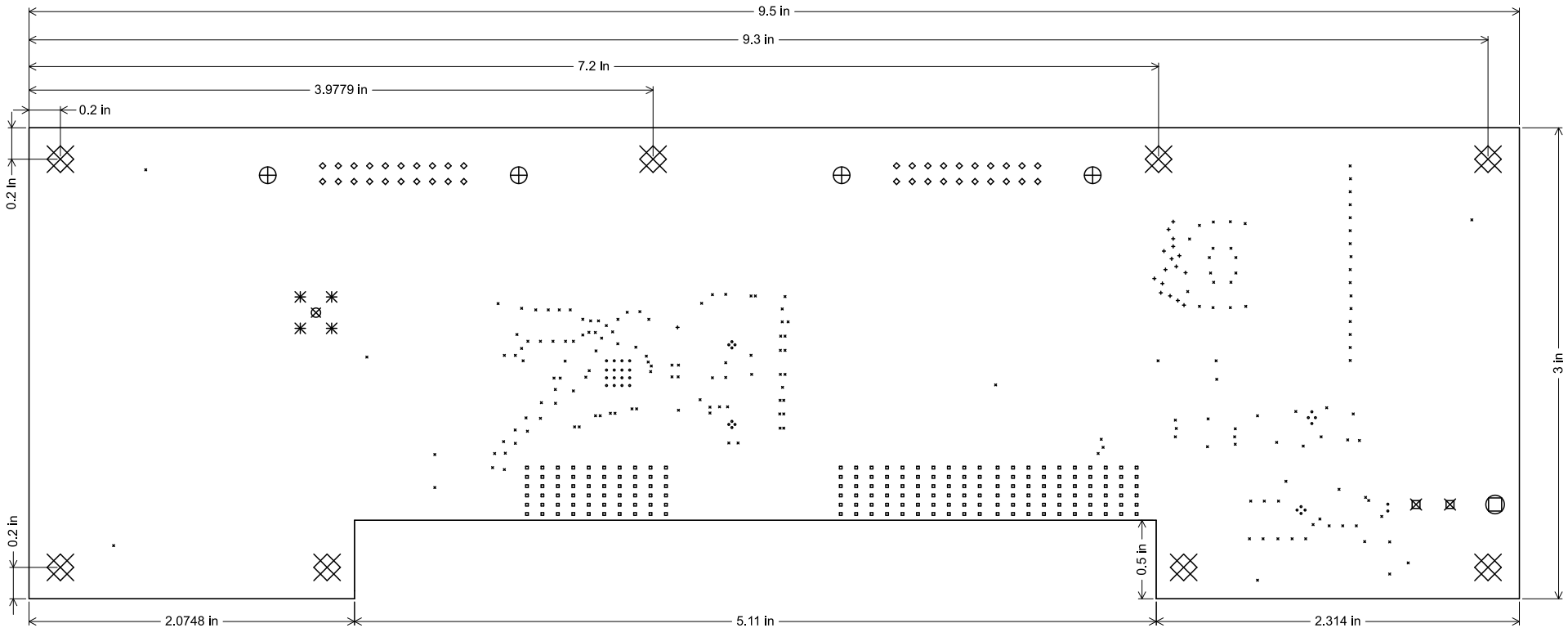
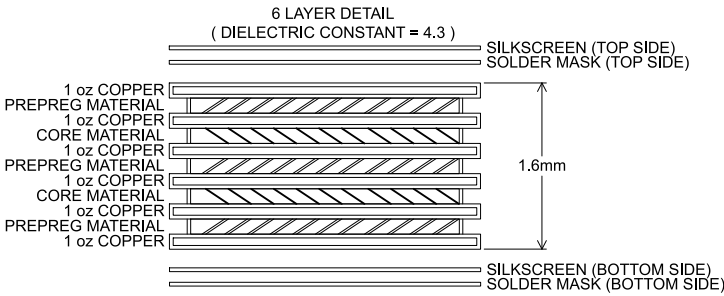
4 In3 Sig

5 In4 GND

6 Bot Sig

Edge coupled microstrip .17/.15 mm 100 Ohm differential 10%

THESE NUMBERS ARE FOR REFERENCE ONLY. FINAL WIDTHS AND SEPERATIONS TO ACHIEVE
100 OHMS DIFFERENTIAL IMPEDANCE SHALL BE THE RESPONSIBILITY OF THE BOARD FABRICATOR.



DRILL DRAWING

SK-2601-52

REV-1

Drill Map:

- 0.300mm / 0.0118" (190 holes)
- 0.300mm / 0.0118" (34 holes)
- 0.500mm / 0.0197" (17 holes)
- 0.510mm / 0.0240" (180 holes)
- 0.970mm / 0.0382" (40 holes)
- ⊗ 1.500mm / 0.0591" (1 hole)
- ⊗ 1.700mm / 0.0669" (4 holes)
- ⊗ 1.800mm / 0.0709" (2 holes)
- ⊗ 4.300mm / 0.1693" (8 holes)
- ⊗ 2.690mm / 0.1059" (4 holes) (not plated)
- 3.000mm / 0.1181" (1 hole) (not plated)

ALL HOLES 0.610MM / 0.0240" PRESS FIT TO -0.84 mil / +1.68 mil

UNLESS OTHERWISE SPECIFIED
ALL DIMENSIONS ARE IN INCHES
DIMENSIONS IN BRACKETS [XXXX] (WHERE PRESENT)
ARE MILLIMETERS AND ARE FOR REFERENCE ONLY
INTERPRET DRAWING AS PER ANSI Y14.5-1994 OR Y14.5-1995

DIMENSIONAL TOLERANCES	ANGULAR TOLERANCE
X = +/- 0.060	±0.5°
X = +/- 0.030	
.XX = +/- 0.015	
.XXX = +/- 0.005	

FINISH
125/



BROOKHAVEN NATIONAL LABORATORY
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Exploring Life's Mysteries,
Protecting The Future



PCB FAB DRAWING
ACM-II AFE Board

CHECKED BY	Joe Mead	
APPROVAL	N/A	N/A
ENGINEER	Joe Mead	
APPROVAL	Danny Padrazo	
SUPERVISOR		
APPROVAL	Bob Lee	
QA APPROVAL	Nicole Magbulo	

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SK-2601-52

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NEXT ASSY:

PROJECT:



THIRD ANGLE
PROJECTION

SCALE:
SEE DWG VIEW

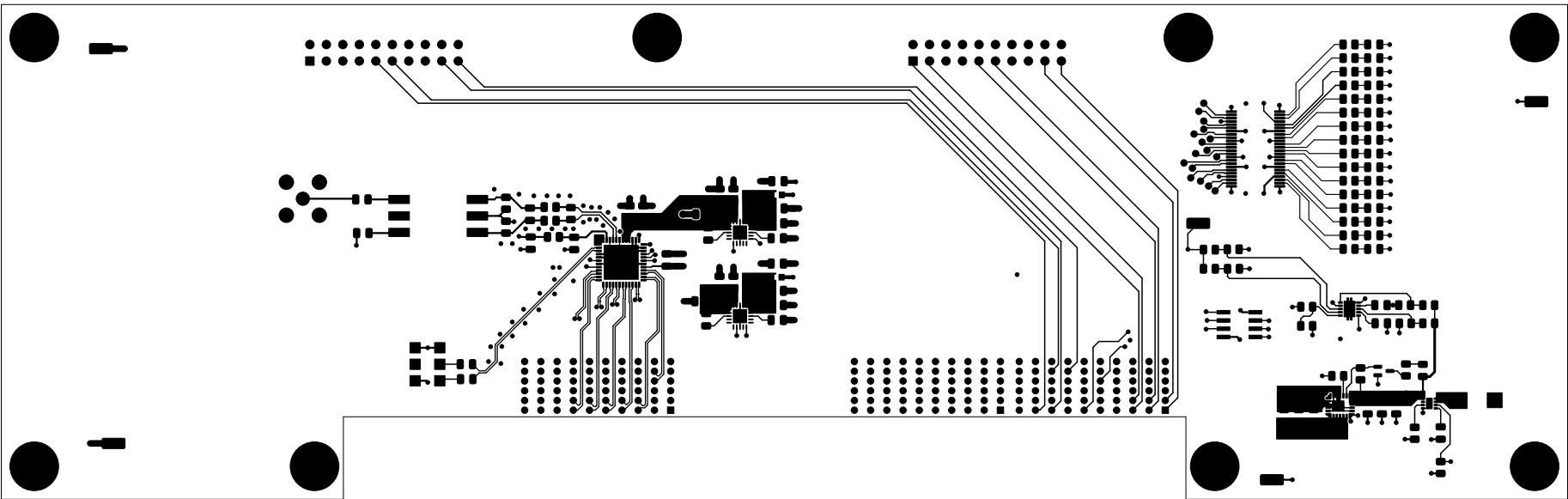
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SHEET SIZE

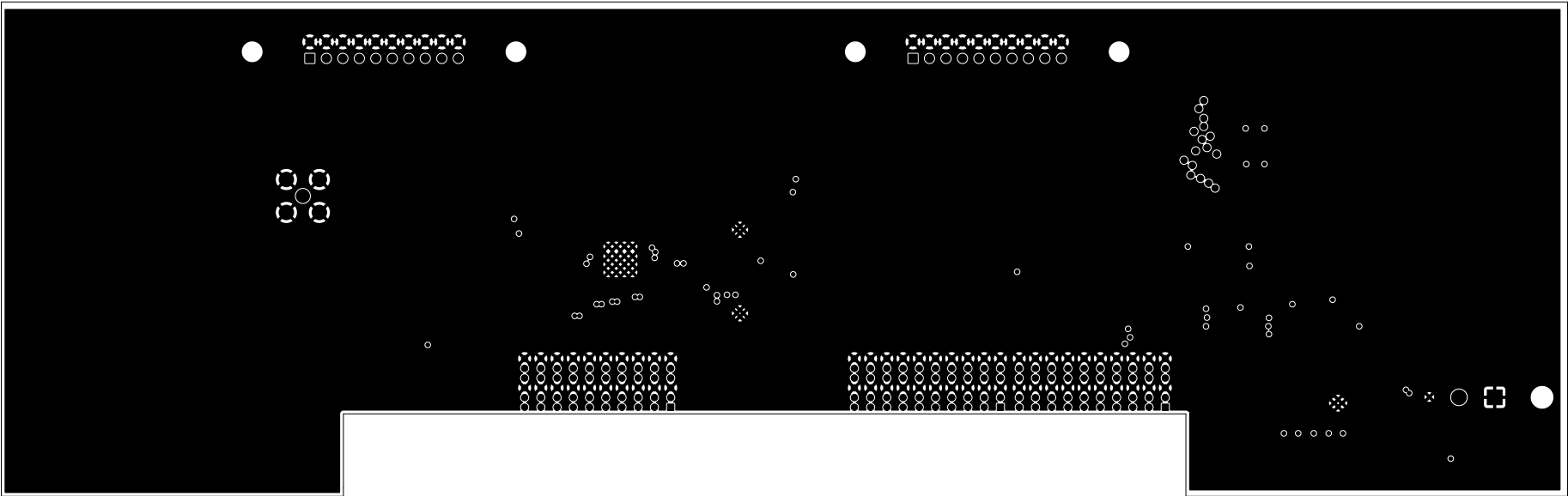
ESHQ
REQ LEVEL

DRAWING/PART NUMBER

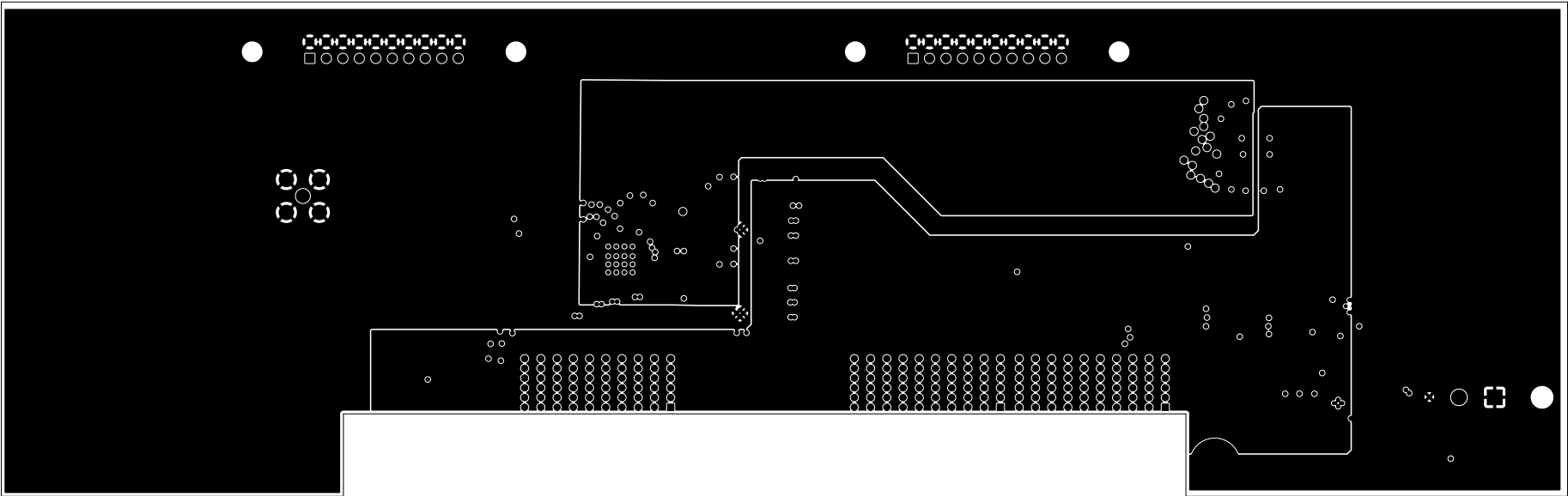
SHEET 1 OF 1



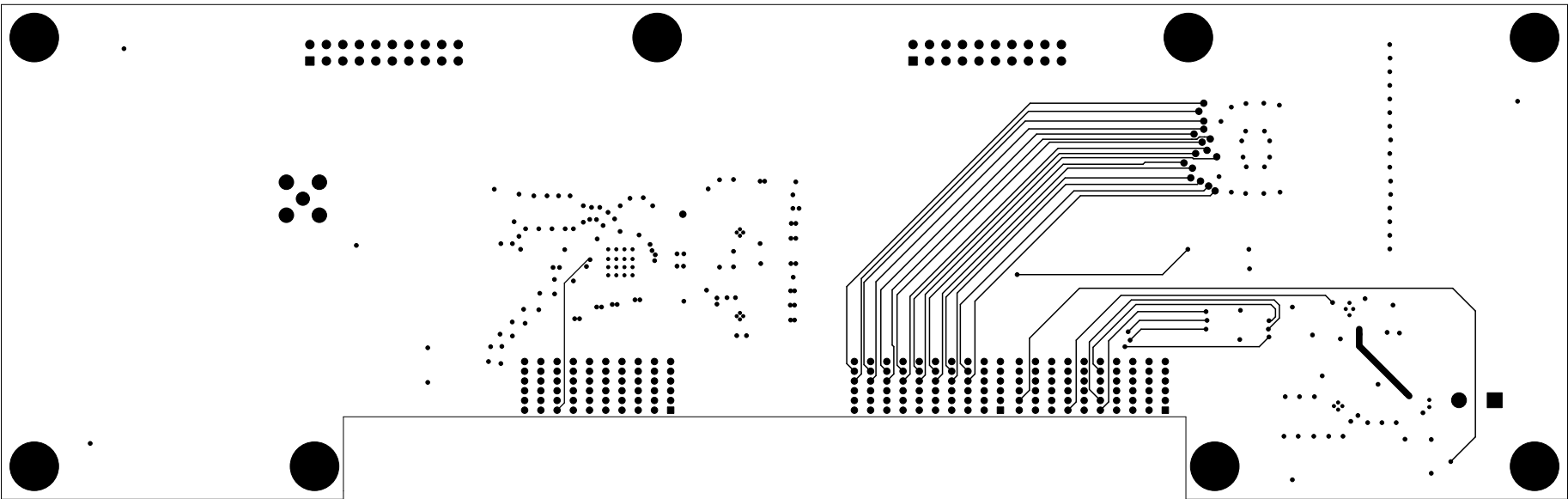
LAYER 1 COMPONENT SIDE
SK-2601-51
REV-1



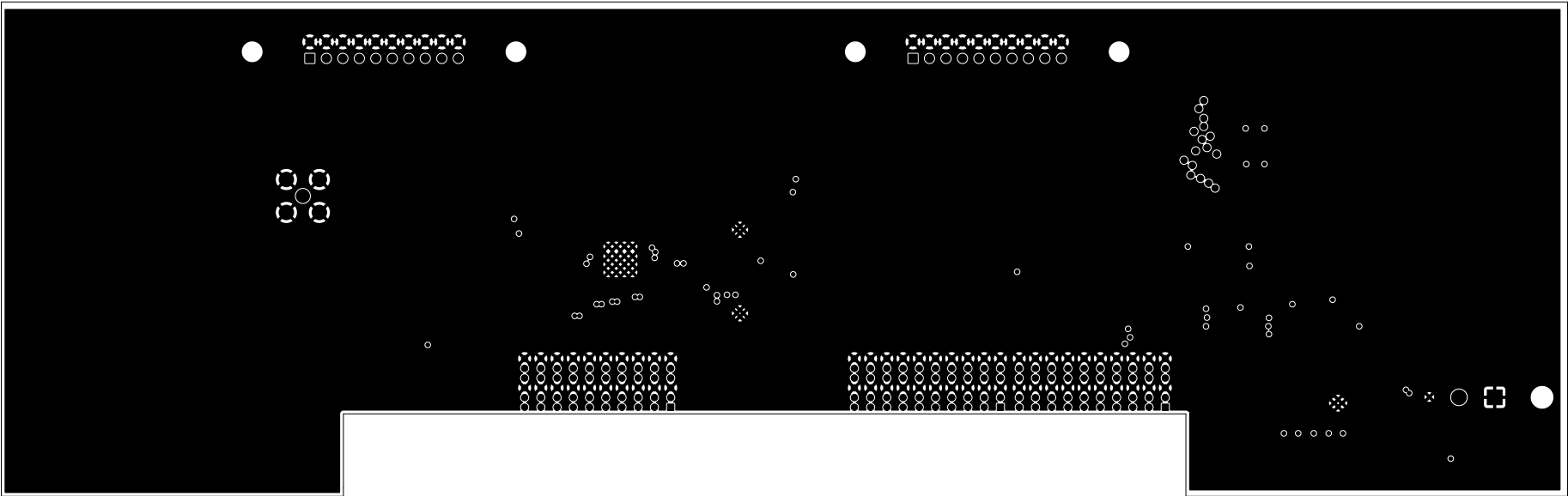
LAYER 2 INTERNAL PLANE
SK-2601-51
REV-1



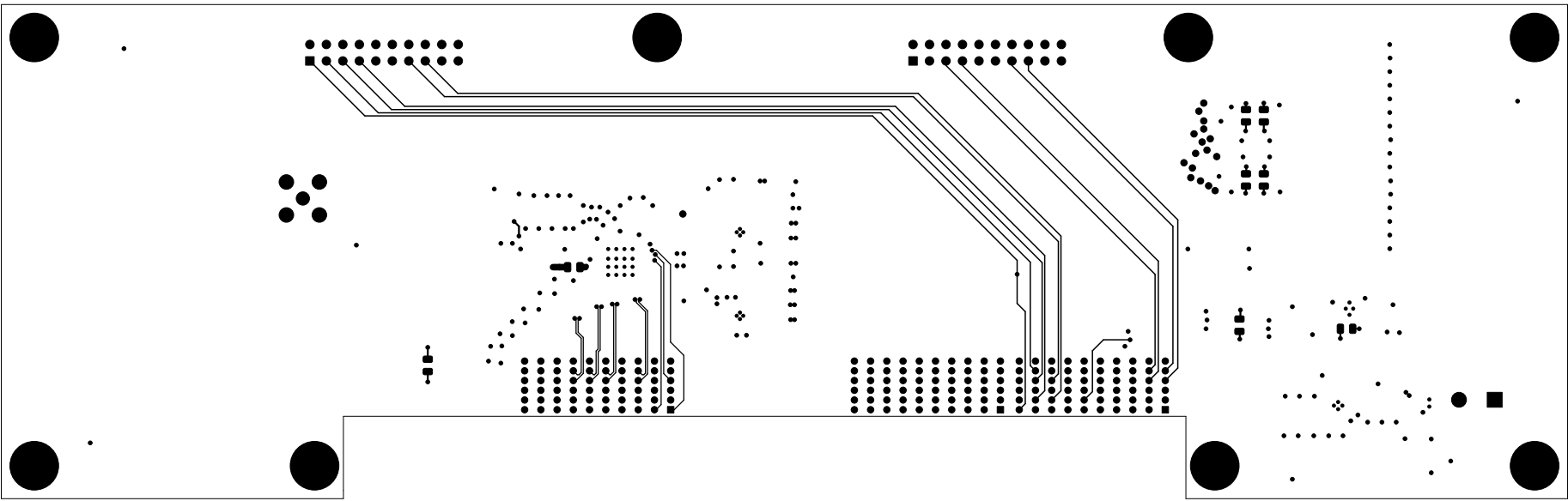
LAYER 3 INTERNAL PLANE
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REV-1



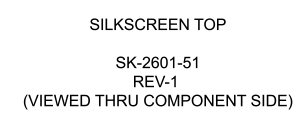
LAYER 4 INTERNAL SIGNAL
SK-2601-51
REV-1



LAYER 5 INTERNAL PLANE
SK-2601-51
REV-1



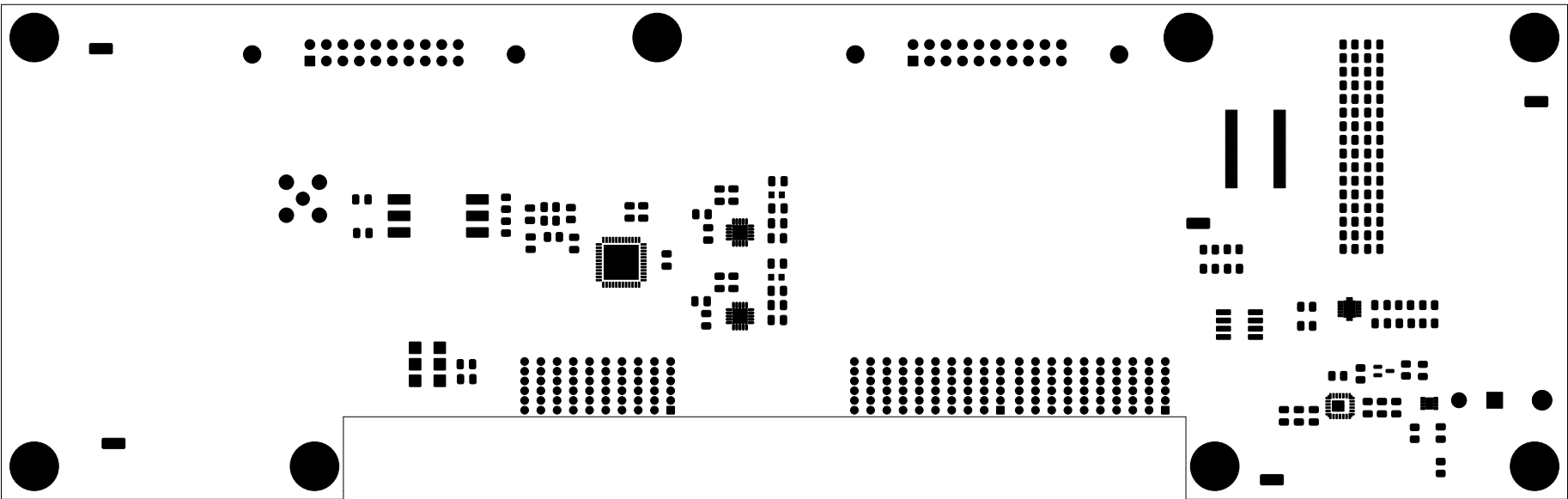
LAYER 6 BOTTOM SIDE
SK-2601-51
REV-1



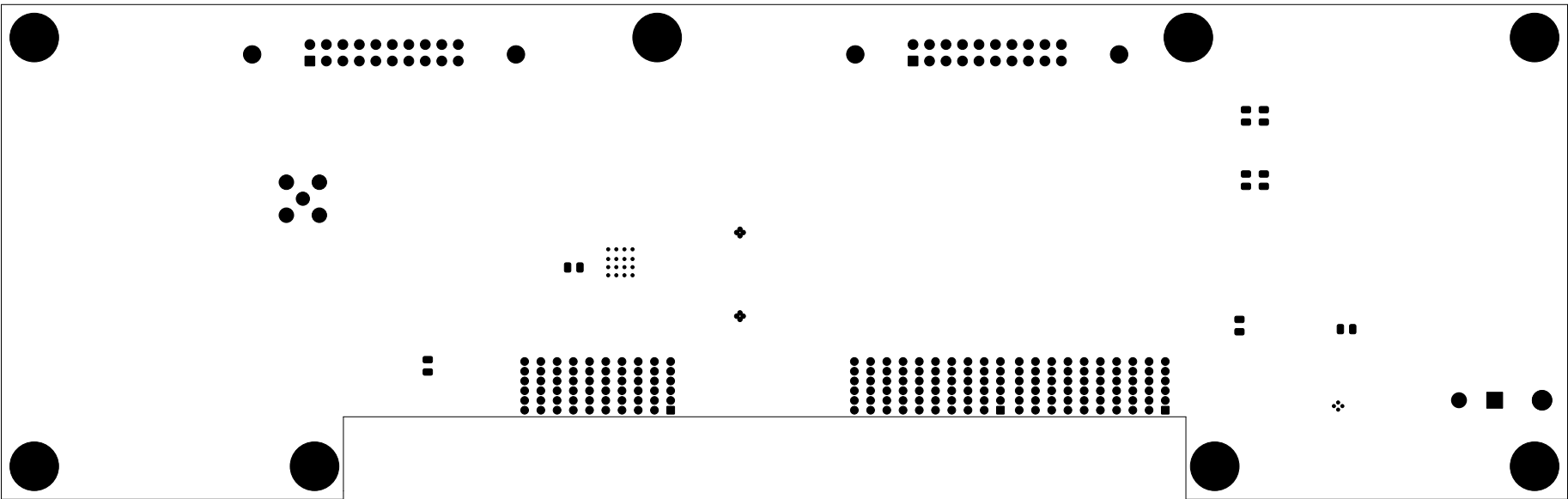
SILKSCREEN TOP
SK-2601-51
REV-1
(VIEWED THRU COMPONENT SIDE)



SILKSCREEN BOTTOM
SK-2601-51
REV-1



SOLDER MASK TOP
SK-2601-51
REV-1



SOLDER MASK BOTTOM
SK-2601-51
REV-1